



Problem solving with theoretical probability

- 1 Two dice are rolled and their values are added together. Assuming I drew the table as efficiently as I could, how many sides are on each dice? Tick **1** correct answer

		dice 1							
		1	2	3	4	5	6	7	8
dice 2	1	2	3	4	5	6	7	8	9
	2	3	4	5	6	7	8	9	10
	3	4	5	6	7	8	9	10	11
	4	5	6	7	8	9	10	11	12

☒ 4 and 8
☐ 4 and 6
☐ 8 and 10
☐ 6 and 6

- 2 The table shows the sample space for when two 8-sided dice are rolled and their values are added together. Which is the most likely outcome? Fill in the blank

		dice 1							
		1	2	3	4	5	6	7	8
dice 2	1	2	3	4	5	6	7	8	9
	2	3	4	5	6	7	8	9	10
	3	4	5	6	7	8	9	10	11
	4	5	6	7	8	9	10	11	12
	5	6	7	8	9	10	11	12	13
	6	7	8	9	10	11	12	13	14
	7	8	9	10	11	12	13	14	15
	8	9	10	11	12	13	14	15	16

9



- 3 The table shows the sample space for when a 6-sided dice and a 4-sided dice are rolled and their values are added together. What is the probability that the dice sum to 5? Tick **1** correct answer

		dice 1			
		1	2	3	4
dice 2	1	2	3	4	5
	2	3	4	5	6
	3	4	5	6	7
	4	5	6	7	8
	5	6	7	8	9
	6	7	8	9	10

- ☐ $\frac{1}{8}$
- ☐ $\frac{5}{8}$
- ☐ $\frac{4}{12}$
- ☒ $\frac{4}{24}$
- ☐ $\frac{5}{24}$

- 4 The tables show the sample space for scenarios where two dice are rolled and their values are added together. Which sample space has the greatest probability of the dice summing to 9? Tick **1** correct answer

		dice 1			
		1	2	3	4
dice 2	1	2	3	4	5
	2	3	4	5	6
	3	4	5	6	7
	4	5	6	7	8

		dice 1					
		1	2	3	4	5	6
dice 2	1	2	3	4	5	6	7
	2	3	4	5	6	7	8
	3	4	5	6	7	8	9
	4	5	6	7	8	9	10
	5	6	7	8	9	10	11
	6	7	8	9	10	11	12

		dice 1							
		1	2	3	4	5	6	7	8
dice 2	1	2	3	4	5	6	7	8	9
	2	3	4	5	6	7	8	9	10
	3	4	5	6	7	8	9	10	11
	4	5	6	7	8	9	10	11	12
	5	6	7	8	9	10	11	12	13
	6	7	8	9	10	11	12	13	14
	7	8	9	10	11	12	13	14	15
	8	9	10	11	12	13	14	15	16

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- 5 The tables show the sample space for scenarios where two dice are rolled and their values are added together. Which sample space has the greatest probability of the dice summing to 7? Tick **1** correct answer

		dice 1			
		1	2	3	4
dice 2	1	2	3	4	5
	2	3	4	5	6
	3	4	5	6	7
	4	5	6	7	8

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		dice 1					
		1	2	3	4	5	6
dice 2	1	2	3	4	5	6	7
	2	3	4	5	6	7	8
	3	4	5	6	7	8	9
	4	5	6	7	8	9	10
	5	6	7	8	9	10	11
	6	7	8	9	10	11	12

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		dice 1							
		1	2	3	4	5	6	7	8
dice 2	1	2	3	4	5	6	7	8	9
	2	3	4	5	6	7	8	9	10
	3	4	5	6	7	8	9	10	11
	4	5	6	7	8	9	10	11	12
	5	6	7	8	9	10	11	12	13
	6	7	8	9	10	11	12	13	14
	7	8	9	10	11	12	13	14	15
	8	9	10	11	12	13	14	15	16

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- 6 The tables show the sample space for scenarios where two dice are rolled and their values are added together. Which sample space has the greatest probability of the dice summing to 6? Tick **1** correct answer

		dice 1			
		1	2	3	4
dice 2	1	2	3	4	5
	2	3	4	5	6
	3	4	5	6	7
	4	5	6	7	8

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		dice 1					
		1	2	3	4	5	6
dice 2	1	2	3	4	5	6	7
	2	3	4	5	6	7	8
	3	4	5	6	7	8	9
	4	5	6	7	8	9	10
	5	6	7	8	9	10	11
	6	7	8	9	10	11	12

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		dice 1							
		1	2	3	4	5	6	7	8
dice 2	1	2	3	4	5	6	7	8	9
	2	3	4	5	6	7	8	9	10
	3	4	5	6	7	8	9	10	11
	4	5	6	7	8	9	10	11	12
	5	6	7	8	9	10	11	12	13
	6	7	8	9	10	11	12	13	14
	7	8	9	10	11	12	13	14	15
	8	9	10	11	12	13	14	15	16

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